Name: Aditya Wadhawan

Roll No: 2401010023

Experiment: 7

Questions:

Question 1: Reverse Words in a String.
Problem:
Write a function that takes a string and reverses each word individually, while keeping the word order the same.

Question 2: Find Duplicate Elements in an Array.
Problem:
Write a function to find and return all duplicate elements in a given array.

Question 3: Count Character Frequency.
Problem:
Write a function that counts the frequency of each character in a string and returns the result as an object.

Question 4: Merge Two Objects.
Problem:
Write a function to merge two objects. If a key exists in both objects, sum their values.

Question 5: Flatten Nested Array (One Level).
Problem:
Write a function to flatten a nested array by one level only.

Question 6: Find the Longest Word in a String.
Problem:
Write a function that returns the longest word from a given sentence.

Question 7: Group Anagrams.
Problem:
Write a function that groups anagrams together from a given array of strings.

Question 8: Remove Falsy Values from Array.
Problem:
Write a function to remove all falsy values from an array.
Falsy values include: false, 0, "", null, undefined, NaN

Question 9: Convert Array to Object.
Problem:
Write a function to convert an array of key-value pairs into an object.

Question 10: Find Missing Number.
Problem:
Given an array containing numbers from 1 to n with one number missing, write a function to find the missing number.

SOLUTION:

CODE: [Index.html]

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <title>Web Assignment</title>

   <link
href="https://fonts.googleapis.com/css2?family=Orbitron:wght@500&family=Poppins:wght@300;400& display=swap"
rel="stylesheet">

   <link rel="stylesheet" href="style.css">

</head>

 <body>

 <div class="card">

   <h1>JS Problem Solver</h1>

   <select id="question">

      <option value="1">Reverse Words</option>

      <option value="2">Find Duplicates</option>
```

```
        <option value="3">Character Frequency</option>
        <option value="4">Merge Objects</option>
        <option value="5">Flatten Array</option>
        <option value="6">Longest Word</option>
        <option value="7">Group Anagrams</option>
        <option value="8">Remove Falsy</option>
        <option value="9">Array to Object</option>
        <option value="10">Find Missing Number</option>
    </select>
    <input type="text" id="input1" placeholder="Enter input">
    <input type="text" id="input2" placeholder="Second input (if needed)">
    <button onclick="solve()">Solve</button>
    <div id="result"></div>
</div>
<script src="script.js"></script>
</body>
</html>
```

[Style.css]:

```
body {     margin: 0;     height:
100vh;     display: flex;
justify-content: center;
align-items: center;
background-color: #b7ccd1;
```

```
font-family: 'Poppins', sans-
serif;
}

.card { background-color: #ececec;
padding: 45px 40px; border-radius: 20px;
width: 380px; text-align: center; box-
shadow: 0 12px 30px rgba(0,0,0,0.18);
}

h1 { font-family: 'Orbitron', sans-
serif; color: #118c8f; font-size:
26px; margin-bottom: 25px;
}

select, input { width:
100%; padding: 12px;
border-radius: 12px;
border: 2px solid #ff5a5a;
margin-bottom: 15px;
font-size: 14px; outline:
none;
}
```

```
button {      background: linear-gradient(90deg, #ff7a6b,
#ff5a5a);       border:  none;       padding:  13px  30px;
border-radius: 30px;      color: white;      font-size: 14px;
cursor: pointer;
}

#result {      margin-
top:  20px;       font-
size: 14px;
}
```

[Script.js]:

```
function solve() {    let q =
document.getElementById("question").value;    let
input1 = document.getElementById("input1").value;    let
input2 = document.getElementById("input2").value;    let
result;    switch (q) {
    // Q.1 Reverse Words
    case "1":
      result = input1.split(" ").map(w => w.split("").reverse().join("")).join(" ");
      break;

    // Q.2 Find Duplicates
case "2":
```

```
    let arr = input1.split(",").map(Number);        result =
arr.filter((item, index) => arr.indexOf(item) !== index);        result =
[...new Set(result)];        break;


  // Q.3 Character Frequency
  case "3":
    let freq = {};        for (let ch of input1) freq[ch]
= (freq[ch] || 0) + 1;        result = freq;        break;


  // Q.4 Merge Objects
case "4":
    let obj1 = parseObj(input1);
    let obj2 = parseObj(input2);
result = { ...obj1 };        for (let k in obj2)
result[k] = (result[k] || 0) + obj2[k];
break;


  // Q.5 Flatten Array
case "5":
    let flatArr = eval("[" + input1 + "]");
result = flatArr.flat(1);        break;


  // Q.6 Longest Word
  case "6":
```

```
    result = input1.split(" ").reduce((a, b) => a.length > b.length ? a : b, "");

    break;


  // Q.7 Group Anagrams

  case "7":

    let words = input1.split(",");

let map = {};      words.forEach(w => {

let key = w.split("").sort().join("");

if (!map[key]) map[key] = [];

map[key].push(w);


    });

    result = Object.values(map);

    break;


  // Q.8 Remove Falsy

  case "8":

    let arr2 = input1.split(",");

    result = arr2.filter(v => v && v !== "0" && v !== "false" && v !== "null" && v !== "undefined"
&& v !== "NaN");

    break;


  // Q.9 Array to Object

case "9":
```

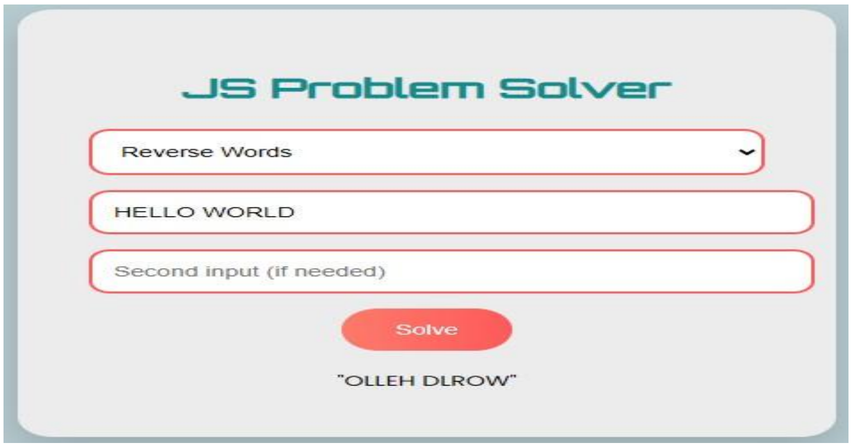
```
        let pairs = input1.split(",");
result = {};          pairs.forEach(p
=> {              let [k, v] =
p.split(":");              result[k] = v;
        });
break;

      // Q.10 Missing Number
      case "10":
        let nums = input1.split(",").map(Number);
        let n = nums.length + 1;          let sum = n *
(n + 1) / 2;          let actual = nums.reduce((a, b)
=> a + b, 0);          result = sum - actual;
break;
  }
  document.getElementById("result").innerText = JSON.stringify(result);
}
function parseObj(str) {
  let        obj        =        {};
str.split(",").forEach(p   =>   {
let    [k,    v]    =    p.split(":");
obj[k] = Number(v);
  });
  return obj;
```

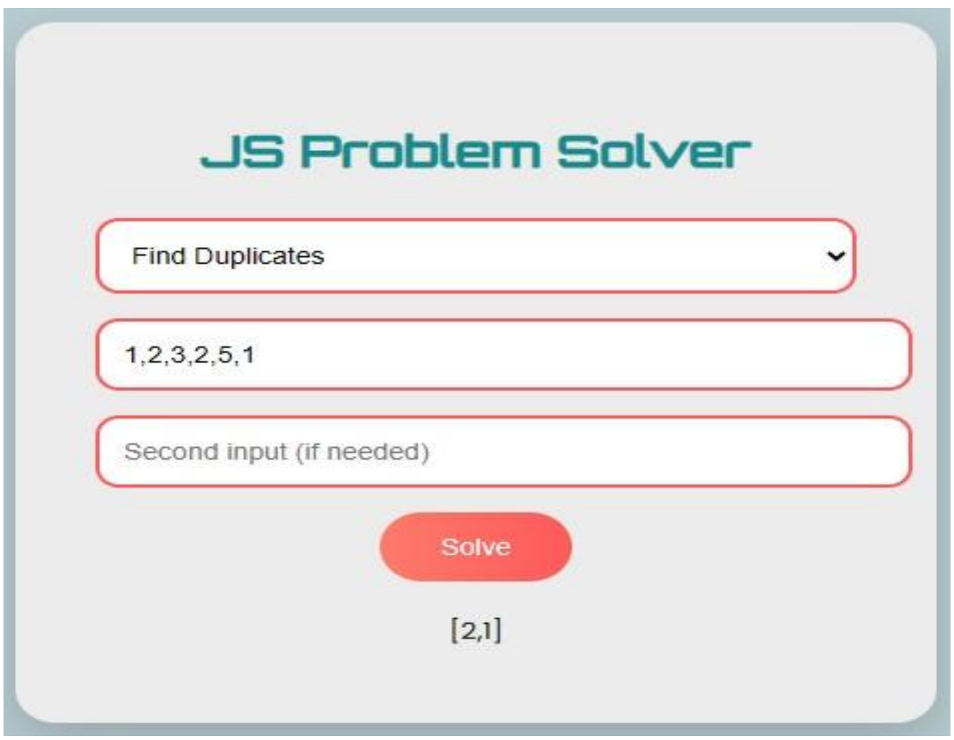
}

[OUTPUT]:

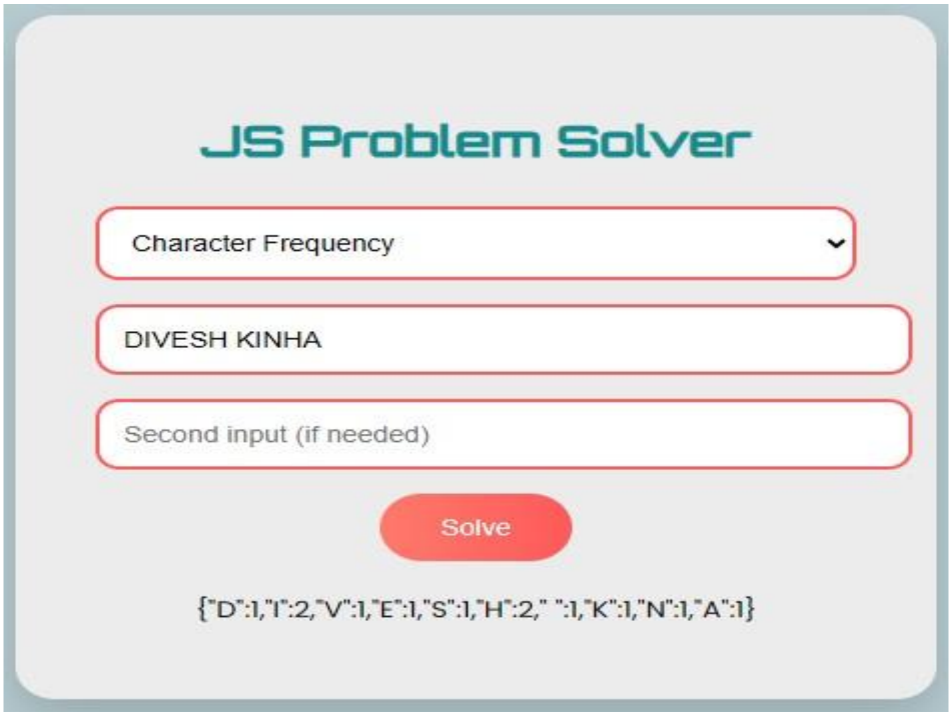
Sol 1: Reverse Word.



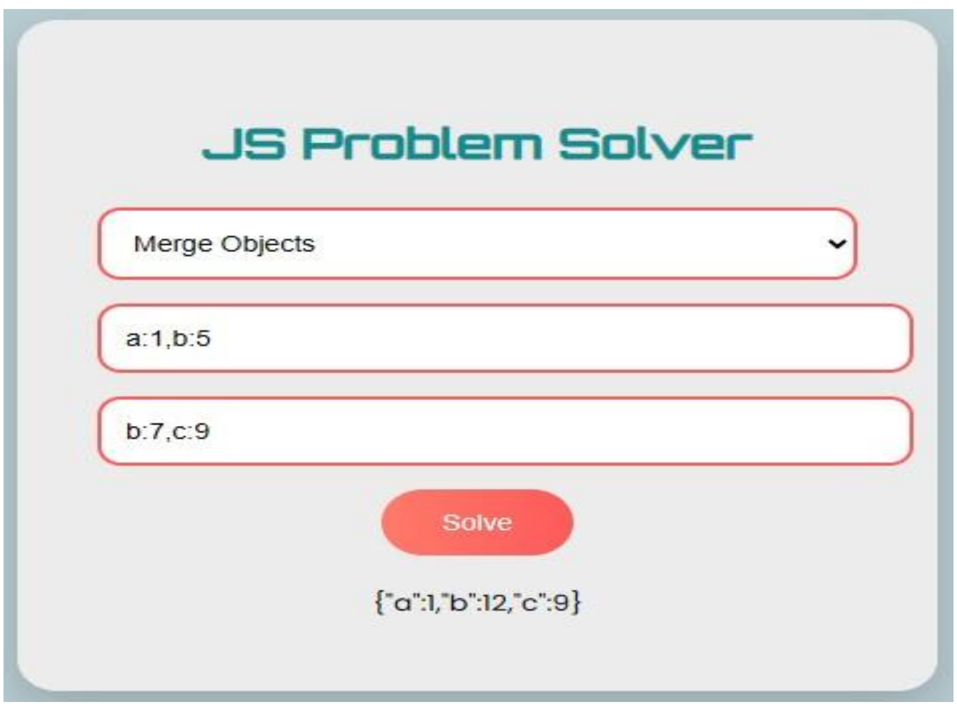
Sol 2: Find Duplicates.



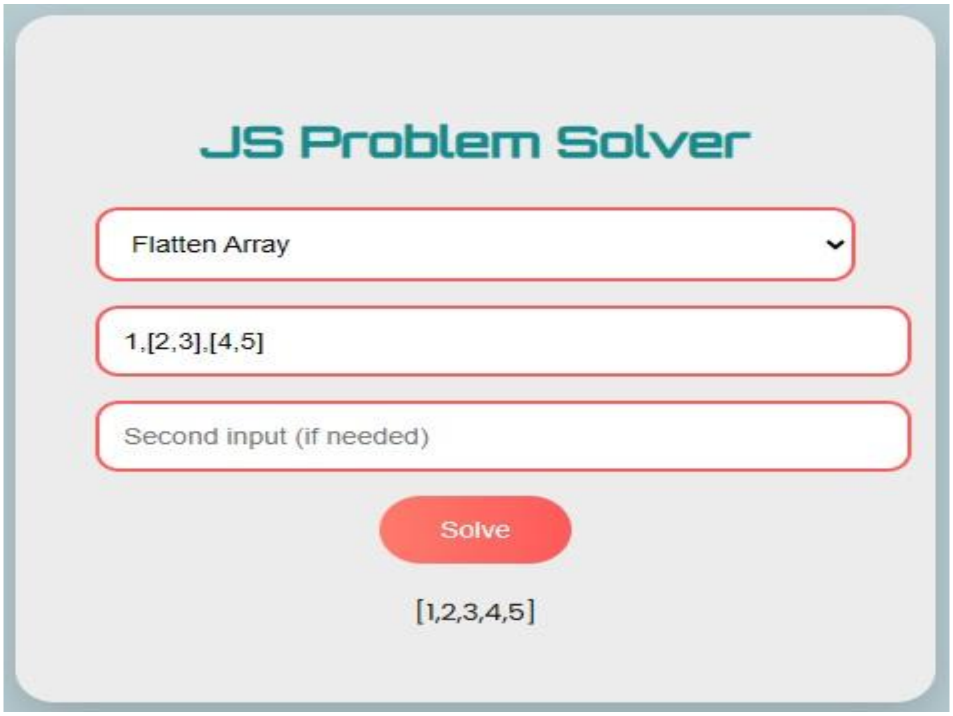
Sol 3: Character Frequency.



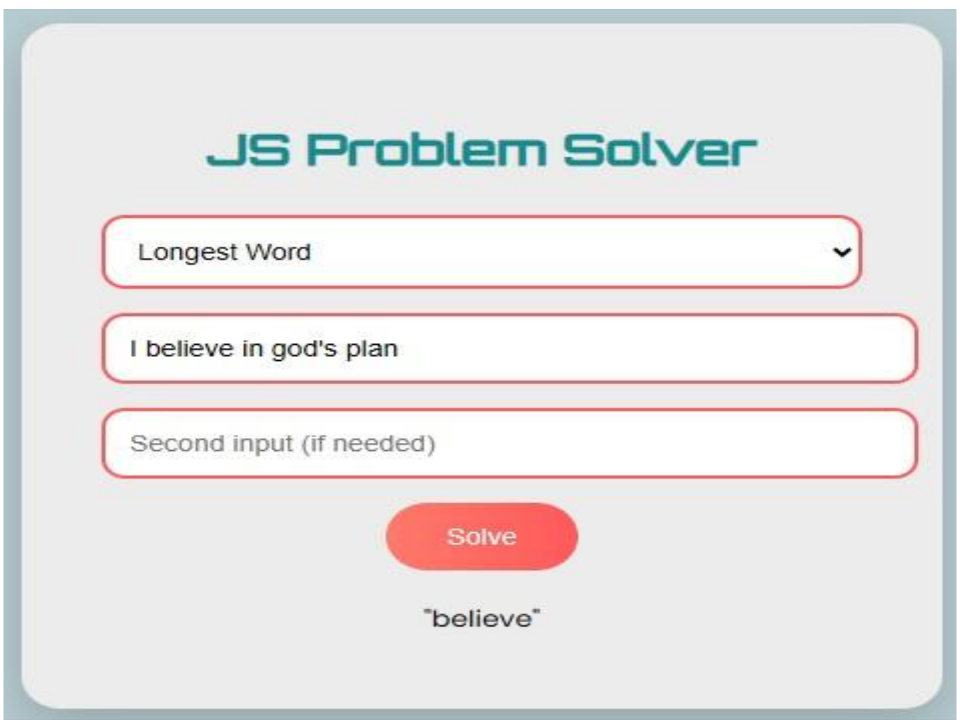
Sol 4: Merge Objects.



Sol 5: Flatten Array.

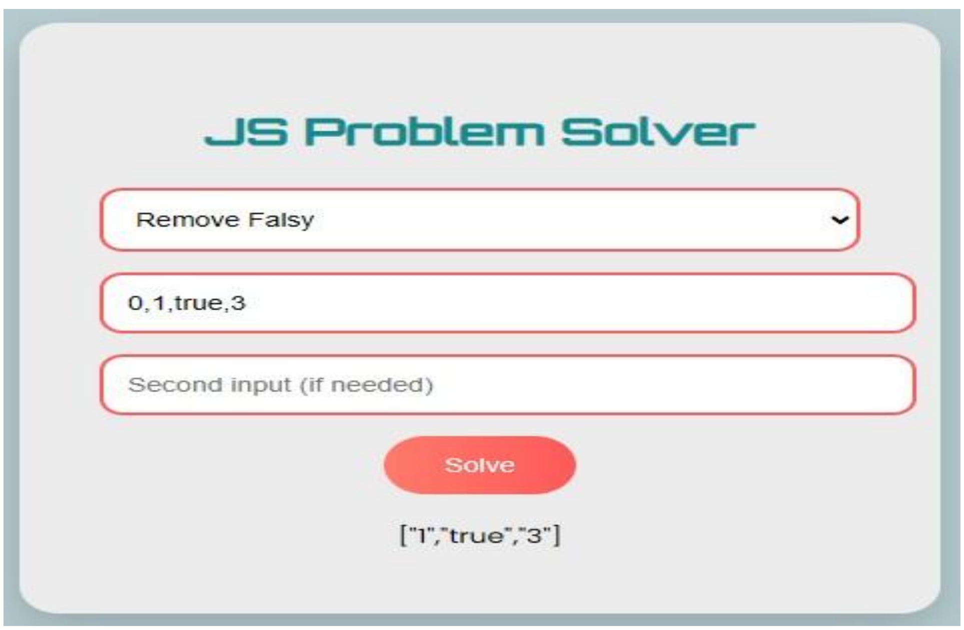


Sol 6: Longest Word.

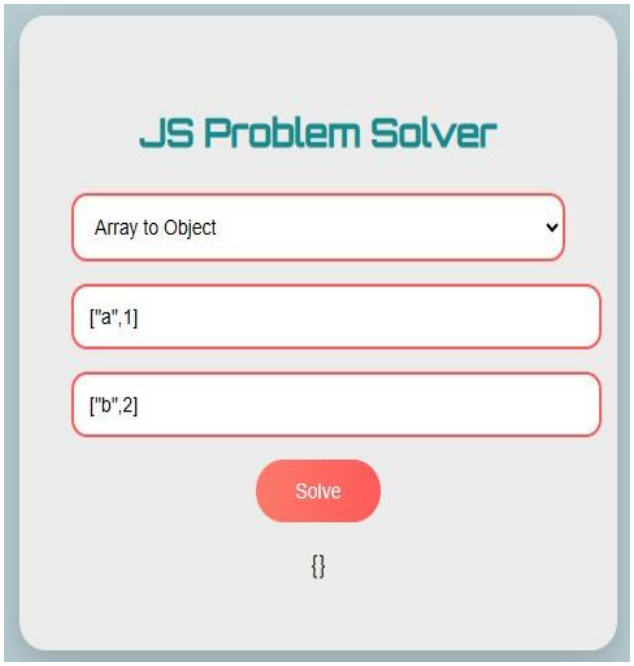


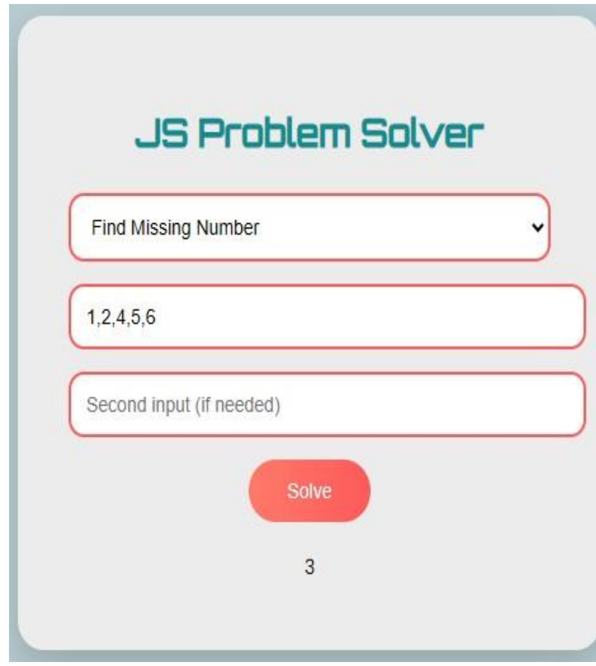
Sol 7: Group Anagrams.

Sol 8: Remove falsy.



Sol 9: Array to Object. **Sol 10: Find Missing No.**


JS Problem Solver
Array to Object
["a",1]
["b",2]
Solve
{}


JS Problem Solver
Find Missing Number
1,2,4,5,6
Second input (if needed)
Solve
3